

Data Flow Diagram

Project: PocketSmart AI – Your Smart Budget & Recommendation Assistant

1. Overall Data Flow

The PocketSmart AI data flow starts with user input, passes through the FastAPI backend and Gemini AI recommendation layer, and returns the generated recommendation to the user. Recommendation information can also be stored for history and later viewing.

Step	Data Flow
1. User Input	User enters budget and planning requirements through the application interface.
2. Frontend	The frontend collects the form data and sends the request to the backend.
3. FastAPI Backend	The backend receives and validates the request and routes it to the appropriate planner.
4. Gemini AI	The relevant planning information is sent to Gemini AI to generate a recommendation.
5. Recommendation	The generated recommendation is returned to the backend and prepared for display.
6. User Output	The recommendation is displayed to the user through the application interface.
7. History	Generated recommendation information can be stored and later retrieved through Recommendation History

2. Planner-Specific Flow

Home Planner: Budget and home requirements → FastAPI → Gemini AI → Home recommendations.

Party Planner: Budget, guest count, event type and venue → FastAPI → Gemini AI → Party recommendations.

Jewelry Planner: Budget, occasion, style and optional outfit image → FastAPI → Gemini AI → Jewelry recommendations.

3. Simplified Flow

User → Frontend → FastAPI Backend → Gemini AI → Recommendation → Frontend → Recommendation History